Sabiq Islam

(512) 550-4979 | sabiqislam@utexas.edu

EDUCATION

Bachelor of Science, Electrical and Computer Engineering Honors, 2022 - 2026
The University of Texas at Austin

Doctorate of Philosophy (In Progress), Electrical and Computer Engineering, 2026 - Present
The University of Texas at Austin

EXTRACURRICULAR ACTIVITIES

- CDCM MRSEC Volunteer - Facilitated hands-on engineering and materials science activities for K–12 students through Introduce a Girl to Engineering Day and UT Engineering school visits.
- Undergraduate Research Mentor -Guided an undergraduate researcher in density functional theory (DFT) workflows, scientific computing, and materials-discovery methods, including simulation setup and interpretation of first-principles results.

RESEARCH EXPERIENCE

Research Assistant, Integrated Nano-Computing Lab of Dr. Jean Anne Incorvia, The University of Texas at Austin, 12/26/2023-Present

- Conducting independent research on novel materials for Magnetic Tunnel Junctions (MTJs) and neuromorphic computing applications.
- Led project collaboration in joint development program (JDP) with TSMC to develop accelerated materials discovery paradigms for new material platforms for magnetoresistive random access memory (MRAM) devices by leveraging a combination of natural language processing (NLP) and first principles-based screening to predict new materials based on correlations in materials science literature.
- Leveraged DFT in order to simulate band structures and quantum transport for materials in MTJ heterostructures.
- Integrated spin-polarized DFT relaxations in GPAW with ASE-driven LAMMPS workflows for geometry optimization of tunnel-junction stacks.
- Fabricated MTJ stacks of different barrier materials using AJA magnetron sputtering system.
- Used MicroSense Vibrating-Sample Magnetometer (VSM) to study and characterize magnetic properties of ferromagnetic materials such as magnetic anisotropy.
- Determined sputter deposition growth rates of thin films by measuring film thicknesses with step-height analysis using a Park Systems NX10 Atomic Force Microscope (AFM).
- Trained NLP models such as Word2Vec, BERT, and LLaMa 3 on a large corpus of scientific literature abstracts in order to find novel magnetic tunnel barrier materials.
- Utilized compute nodes from Texas Advanced Computing Center (TACC) to operate some of the world's most powerful computing resources for machine learning (ML) research.

Research Intern, ECE Next REU Program, The University of Texas at Austin, 05/27/2025 - 07/25/2025

- Selected as one of eleven participants in paid research internship.
- Led project on first principles-based analysis of novel MTJ devices, sponsored by TSMC.
- Used Birch clustering and Principal Component Analysis on DFT-derived band structure features for materials discovery

Research Intern, ECE Next REU Program, The University of Texas at Austin, 05/28/2024 - 07/26/2024

- Selected as one of five participants in paid research internship focusing on cutting edge semiconductor research.
- Led project on leveraging NLP models to discover novel materials to be used in MTJs sponsored by TSMC.

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Research Intern, Sustainable Nano-Engineering for Water Treatment Group of Dr. Navid Saleh, The University of Texas at Austin, 06/01/2019-08/15/2021

- Built Ceramic Water Filters (CWF's) coated with Silver Nanoparticles under the supervision of Ph.D. students.

PATENTS AND DISCLOSURES

- ***Islam, S. Z.***, Rogers, W. S., Incorvia, J. A. C. "Enhanced Magnetoresistance via Nitride Layer Dusting in Magnetic Tunnel Junctions (MTJs)" U.S. Provisional Patent Application, filed May 2025.

PRESENTATIONS

- ***Islam, S.***, Rogers, W., Hu, C.-Y., Song, M., Bao, X., Sayed, S., Incorvia, J. A. C. "Predicting Novel Tunnel Barrier Materials for Magnetic Tunnel Junctions using Language Models and First-Principles Screening," ***The 17th MRAM Global Innovation Forum***, December 2025, San Francisco, CA (Poster Presentation)
- ***Islam, S.***, Rogers, W. S., Hu, C.-Y., Song, M., Bao, X., Sayed, S., Incorvia, J. A. C. "Literature-Guided First Principles Investigation of Nitride Interfacial Layers for Magnetic Tunnel Junctions with High Tunnel Magnetoresistance and Low Resistance Area Product," ***The 10th Annual 2025 International Conference on Rebooting Computing (ICRC)***, December 2025, San Diego, CA (Oral Presentation)
- ***Islam, S.***, Rogers, W. S., Hu, C.-Y., Song, M., Bao, X., Sayed, S., Incorvia, J. A. C. "Unsupervised Learning-Guided First-Principles Analysis of Magnetic Tunnel Junctions with Nitride Dustings Exhibiting Large Magnetoresistance and Low Resistance-Area Product," ***The 2025 IEEE Around-the-Clock Around-the-Globe Magnetics Conference (AtC-AtG)***, November 2025 (Oral Presentation)
- ***Islam, S.***, Rogers, W., Hu, C.-Y., Song, M., Bao, X., Sayed, S., Incorvia, J. A. C. "Predicting Novel Tunnel Barrier Materials for Magnetic Tunnel Junctions using Language Models and First-Principles Screening," ***The 70th Annual Conference on Magnetism and Magnetic Materials (MMM)***, October 2025, Palm Beach, FL (Oral Presentation)
- ***Islam, S.***, Rogers, W., Incorvia, J. A., Sayed, S., "Enhanced Tunnel Magnetoresistance and Lower Resistance Magnetic Tunnel Junctions Prediction using Language Models and First-Principles Screening," ***ECE Next REU Poster Symposium***, July 2025, Austin, TX (Poster Presentation)
- ***Islam, S.***, Rogers, V., Incorvia, J. A., Sayed, S., "Materials Predictions For Nanomagnetic Devices Using Large Language Models And First Principles Screening," ***ECE Next REU Poster Symposium***, July 2024, Austin, TX (Poster Presentation)

PUBLICATIONS

- ***S. Islam***, W. S. Rogers, C.-Y. Hu, M. Song, X. Bao, S. Sayed, J. A. C. Incorvia" LEAD: Literature Enhanced Ab Initio Discovery of Nitride Dusting Layers for Enhanced Tunnel Magnetoresistance and Lower Resistance Magnetic Tunnel Junctions." ***Adv. Mater.*** (2025): e18241. https://doi.org/10.1002/adma.202518241
- A. Hossain, O. G. Apul, S. Dev, S. Das, ***S. Islam***, C.-Y. Lai, H. Huntington, M. J. Kirisits, S. Umanzor, W.-T. Chen, S. Aggarwal, N. B. Saleh, "A Symbiotic Engineering Approach for Microplastic Remediation in Mariculture Systems," ***ACS ES&T Engineering*** (2022). https://doi.org/10.1021/acsestengg.1c00333
- N. B. Saleh, J. Brown, M. Elliott, J. Carrera, J. Guest, A. Hossain, M. Medina, B. Bzdyra, L. S. Rowles III, ***S. Islam***, "A Structured Framework of Convergent Research to Engineer Appropriate Water Treatment and Sanitation Technologies for Low-income Communities: A U.S. Perspective," ***ACS ES&T Engineering*** (Submitted, 2021).

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- L. S. Rowles III, A. A. Ahmad, ***S. Z. Islam***, D. F. Lawler, N. B. Saleh, "Sustained Ionic Release from Silver Nanoparticles in Varying Water Chemistry: Integrating Navajo Pottery Techniques to Improve Functionality of Ceramic Water Filter," ***8th Sustainable Nanotechnology Organization Conference*** (2019).

TECHNICAL SKILLS

QuantumATK Nanolab, Quantum ESPRESSO, GPAW, LAMMPS, Wannier90, COMSOL Multiphysics, Cadence Virtuoso, Calibre LVS/DRC, LTSPICE, Verilog, KiCad, Python, FEniCS, NLP models (BERT, LLaMA 3, Word2Vec), Keil µVision (ARM Cortex-M), LabVIEW, SolidWorks.

FABRICATION AND CHARACTERIZATION SKILLS

Sputtering, VSM, AFM, Photolithography, Optical Microscopy, Ellipsometry, Diffusion, Etching, Thermal Evaporation, X-Ray Diffraction

RELEVANT COURSEWORK

VLSI CAD and Optimization, Quantum Theory of Electronic Materials, Solid-State Electronic Devices, Integrated Circuit Nano Manufacturing Techniques